LUMI: Data-Driven Environmental Intelligence System for Renewable Energy Decision Support

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PROJECT CONTEXT:

Climate change in the Philippines is intensifying, leading to frequent extreme weather events such as heat waves, strong typhoons, and heavy rainfall. The Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA) indicates that these climate shifts since the mid-20th century are primarily driven by human activities, particularly increased greenhouse gas emissions from land-use changes and fossil fuel combustion. These factors exert mounting pressure on the environment and society, with future implications anticipated as the country remains heavily reliant on fossil fuels for energy, underscoring the urgent need for renewable and sustainable energy solutions.

Lenain (2026) highlighted that the Philippine government is making significant investments in climate adaptation initiatives to mitigate the long-term effects of climate change. Additionally, the country is enhancing its renewable energy contributions as part of global climate mitigation efforts. Nevertheless, the adoption of renewable energy faces challenges, particularly due to fragmented climate and energy-related data that are often presented in complex formats, hindering public understanding and utilization. Although information on climate change is abundant, there is a noticeable scarcity of user-friendly, data-driven tools that effectively illustrate the consequences of fossil fuel use and the feasibility of transitioning to renewable energy sources.

There is a significant amount of climate and energy-related data available, however, there is a lack of systems that effectively transform this data into accessible and actionable insights for users. Existing information sources often present data in ways that are difficult to interpret, limiting the ability of individuals to make informed decisions regarding energy use and sustainability.

To address this gap, this study proposes Lumi, a data-driven environmental intelligence system designed to provide users with accessible insights and predictive analytics. It aims to support users in understanding current energy consumption and demand trends and evaluating suitable renewable energy options through data visualizations, scenario-based simulations, and recommendation mechanisms.

HYPOTHESIS (NULL & ALTERNATIVE)

H_0:

- The system does not significantly improve users’ ability to understand and evaluate renewable energy options.

H_1:

- The system significantly improves users’ ability to understand and evaluate renewable energy options.

PURPOSE AND DESCRIPTION:

Lumi aims to deliver accessible, data-driven information about alternative energy sources, facilitating users' understanding and evaluation of renewable energy in relation to their current usage. Additionally, it seeks to enhance awareness of climate change and greenhouse gas emissions by providing insights on their environmental and societal impacts, thereby encouraging users to make informed and sustainable energy choices.

The system features a user-friendly dashboard that presents climate and energy data, including temperature trends, rainfall patterns, and energy consumption and demand trends, through clear visualizations and AI-assisted interpretations. These indicators help assess the suitability of various renewable energy sources. It has a predictive analytics module for forecasting future energy demand and a renewable energy recommendation engine that suggests sources like solar and wind based on regional conditions. A what-if simulation tool allows users to compare different energy scenarios, while the region selection and energy profile module enables viewing of climate overviews and energy trends for predefined Philippine regions, supported by AI-based recommendations.

The system employs data analytics to process and visualize climate and energy data, transforming complex datasets into insights. It uses rule-based AI for recommendations based on environmental data, alongside a predictive analytics module for estimating future energy demand trends. Together, these technologies enhance decision-making and offer a user-friendly platform for renewable energy evaluation.

GENERAL & SPECIFIC OBJECTIVES:

A. **General Objective:**
   a. To develop a web-based environmental intelligence system that provides accessible, data-driven insights and predictive analytics to help users understand and evaluate renewable energy options.

B. **Specific Objectives:**
   a. To design and develop a dashboard that visualizes climate patterns and energy trends interpretation;
   b. To develop a rule-based recommendation engine for identifying suitable renewable energy sources;
   c. To implement a predictive analytics module for forecasting energy demand trends;
   d. To create a simple what-if simulation tool for comparing renewable energy scenarios;
   e. To develop a user-friendly interface that supports data interpretation and decision-making.

SCOPE & LIMITATIONS:

This study evaluates Lumi, a web-based environmental intelligence system designed to assist users in assessing renewable energy options. It combines publicly available climate and energy data, like temperature and rainfall patterns with interactive visualizations. Key features include a climate and energy dashboard, predictive analytics for energy demand, a rule-based recommendation engine for renewable sources (solar, wind, hydro), and a what-if simulation tool. Additionally, it offers a region selection module for localized insights within the Philippines. Lumi targets students, households, and general users, focusing on user-friendly outputs to improve decision-making in renewable energy selection.

The study has several limitations: it relies on publicly available datasets, affecting the accuracy of results. The predictive analytics module uses basic statistical methods instead of advanced machine learning, which may limit prediction precision. The recommendation engine is rule-based and may overlook real-world variables. It focuses on specific renewable sources such as solar, wind, and hydro, excluding other technologies. Additionally, the what-if simulation tool simplifies comparisons without detailed financial or technical considerations. The system serves informational purposes and should not replace professional energy planning or policy-making.

SIGNIFICANCE OF THE STUDY:

This study emphasizes the importance of a data-driven approach to enhance awareness and understanding of renewable energy and climate-related issues. By converting complex climate and energy data into accessible visual insights, the system assists users in making informed decisions about energy use and sustainability.

For students and educators, it serves as an educational resource that facilitates learning about climate change and renewable energy technologies. For households and community members, it offers practical insights to promote awareness and foster the adoption of sustainable energy practices.

Government agencies and policymakers can utilize the system as a decision-support tool to grasp regional climate and energy trends. By organizing and interpreting data effectively, the system aids in crafting informed strategies for promoting renewable energy, initiating sustainability initiatives, and enhancing public awareness programs.

For researchers and developers, the study lays a groundwork for creating advanced environmental intelligence systems that incorporate artificial intelligence and data analytics. It showcases the practical application of rule-based AI and predictive analytics in tackling real-world environmental challenges.

Overall, the study bridges the gap between technical data and public comprehension, empowering individuals and organizations to make environmentally responsible, data-driven decisions while supporting broader sustainability efforts.